CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 5: **Life Processes** | Quick revision, one page

Nutrition in plants

- Photosynthesis: $6\,CO_2 + 6\,H_2O \xrightarrow{\text{chlorophyll, sunlight}} C_6H_{12}O_6 + 6\,O_2$.
- Steps: absorption of light by chlorophyll; conversion of light energy to chemical energy and splitting of water into H_2 and O_2; reduction of CO_2 to carbohydrates.
- Gases enter/leave through stomata; guard cells open the pore when water flows in. Starch is tested with iodine (blue-black).

Human digestion

- Mouth: salivary amylase (starch $\rightarrow$ sugar). Stomach: HCl, pepsin (protein), mucus protects wall.
- Liver: bile (emulsifies fats; alkaline). Pancreas: amylase, trypsin, lipase. Small intestine: complete digestion; villi absorb.
- Heterotrophic modes: holozoic (humans, Amoeba), saprophytic (fungi), parasitic.

Respiration

- Glucose $\xrightarrow{\text{cytoplasm}}$ pyruvate (3C) $\rightarrow$ (with O_2, mitochondria) $CO_2 + H_2O$ + energy; ATP is the energy currency.
- Anaerobic: yeast gives ethanol + CO_2 + energy; muscles give lactic acid (cramps) + energy.
- Human lungs: nostril $\rightarrow$ pharynx $\rightarrow$ trachea $\rightarrow$ bronchi $\rightarrow$ alveoli (gas exchange); haemoglobin carries O_2; CO_2 is carried in dissolved form.

Transportation

- Heart: 4 chambers (2 atria, 2 ventricles); double circulation; BP about 120/80 mm Hg (systolic/diastolic).
- Arteries: thick walls, carry blood from heart; veins: thinner, valves, carry blood to heart; capillaries: exchange. Blood: plasma, RBC, WBC, platelets.
- Plants: xylem (water, minerals; transpiration pull); phloem (food; translocation uses ATP).

Excretion

- Kidney: nephron = glomerulus + Bowman's capsule + tubule. Filtration $\rightarrow$ selective reabsorption $\rightarrow$ urine collected in pelvis $\rightarrow$ ureter $\rightarrow$ bladder $\rightarrow$ urethra.
- Haemodialysis (artificial kidney) removes urea. Plants: excretion by fallen leaves, bark, gums, resins, transpiration.